

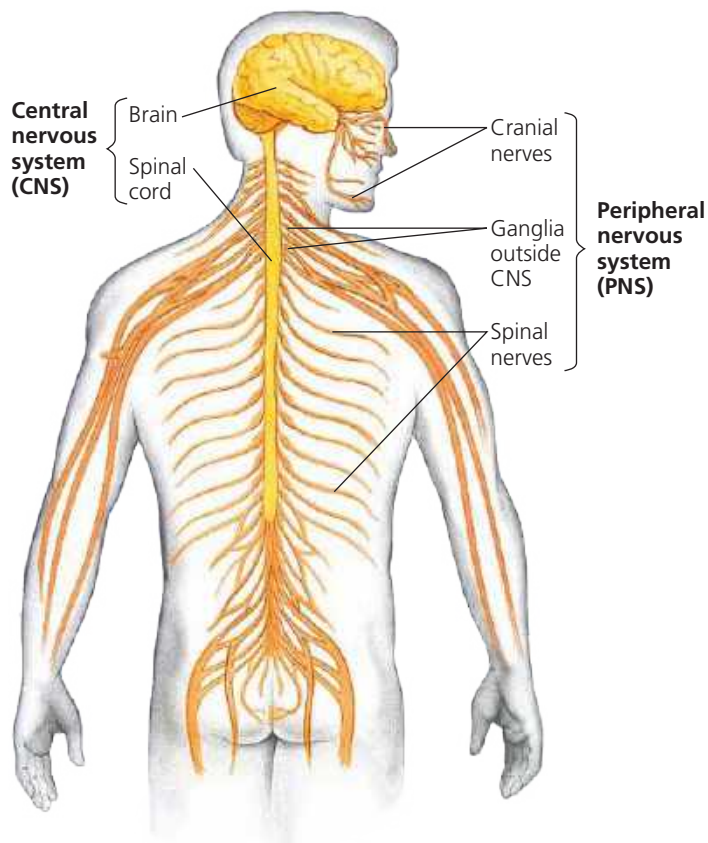
Within an animal group, nervous system organization often correlates with lifestyle. Among the molluscs, for example, sessile and slow-moving species, such as clams and chitons, have relatively simple sense organs and little or no cephalization (**Figure 49.2f**). In contrast, active predatory molluscs, such as octopuses and squids (**Figure 49.2g**), have the most sophisticated nervous systems of any invertebrate. With their large, image-forming eyes and a brain containing millions of neurons, octopuses can learn to discriminate between visual patterns and to perform complex tasks, such as unscrewing a jar to feed on its contents.

In vertebrates such as a salamander (**Figure 49.2h**) or human (**Figure 49.3**), the brain and the spinal cord form the CNS; nerves and ganglia are the key elements of the PNS. Regional specialization is a hallmark of both systems, as we will see throughout this chapter.

Organization of the Vertebrate Nervous System

During embryonic development in vertebrates, the central nervous system develops from the hollow dorsal nerve cord—a hallmark of chordates (see Figure 34.3). The cavity

▼ **Figure 49.3 The vertebrate nervous system.** The central nervous system consists of the brain and spinal cord (yellow). Left-right pairs of cranial nerves, spinal nerves, and ganglia make up most of the peripheral nervous system (dark gold).



of the nerve cord gives rise to the narrow *central canal* of the spinal cord as well as the *ventricles* of the brain. Both the canal and ventricles fill with *cerebrospinal fluid*, which is formed in the brain by filtering arterial blood (**Figure 49.4**). The cerebrospinal fluid supplies the CNS with nutrients and hormones and carries away wastes, circulating through the ventricles and central canal before draining into the veins.

In addition to fluid-filled spaces, the brain and spinal cord have gray and white matter. **Gray matter** is primarily made up of neuron cell bodies. **White matter** consists of bundled axons. In the spinal cord, white matter forms the outer layer, reflecting its role in linking the CNS to sensory and motor neurons of the PNS. In the brain, white matter is predominantly in the interior, where signaling between neurons functions in learning, feeling emotions, processing sensory information, and generating commands.

In vertebrates, the spinal cord runs lengthwise inside the vertebral column, known as the spine. The spinal cord conveys information to and from the brain and generates basic patterns of locomotion. It also acts independently of the brain as part of the simple nerve circuits that produce **reflexes**, the body's automatic responses to certain stimuli.

Reflexes protect the body by providing rapid, involuntary responses to particular stimuli. Reflexes are rapid because sensory information is used to activate motor neurons without the information first having to travel from the spinal cord to the brain and back. If you accidentally put your hand on a hot burner, a reflex jerks your hand back even before your brain processes pain. Similarly, the knee-jerk reflex provides an immediate protective response when you pick up an unexpectedly heavy object. If your legs buckle, the tension across your knees triggers contraction of your thigh muscle (quadriceps), helping you stay upright and support the load.

▼ **Figure 49.4 Ventricles, gray matter, and white matter.**

Ventricles deep in the brain's interior contain cerebrospinal fluid. In the cerebrum, most of the gray matter is on the surface, surrounding the white matter.

